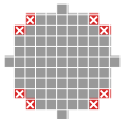
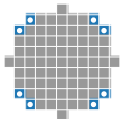


(a) single openings
 $D_5 \circ D_4 \neq D_5$



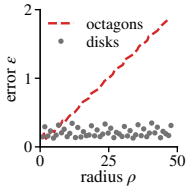
round: **yes**
 granulometry: **no**

(b) sup-closure
 $D_5 \circ D_4 \cup D_5 \circ D_5 = D_5$



round: **yes**
 granulometry: **yes**

(c) one element per size
 octagon chain



round: **no**
 granulometry: **yes**

(d) why (c) is forced
 edge directions, $\|v\| \leq 3$



red: gaps no edge can fill
 $\Rightarrow \varepsilon \gtrsim \rho / (16\pi^2 \delta^2)$